

Problem 1 [20 points]

a) Compute the Laplace transform of

$$f(t) = \begin{cases} 0 & 0 \leq t \leq 1, \\ t & t > 1. \end{cases}$$

b) Use the Laplace transform to find the solution of

$$y'' + y = 2e^t \quad \text{with } y(0) = y'(0) = 0.$$

c) Compute the inverse Laplace transform $\mathcal{L}^{-1}(F)(t)$ of the following function

$$F(s) = \frac{1}{s^2 + 2s + 17}.$$

$$\begin{aligned} \text{a)} \quad \mathcal{L}(f(t))(s) &= \int_0^{\infty} e^{-st} f(t) dt \\ &= \int_0^1 e^{-st} \cdot 0 dt + \int_1^{\infty} e^{-st} t dt \\ &= \int_1^{\infty} e^{-st} t dt \quad \text{Integration by parts: } u v' = uv - \int u'v \\ u &= t \quad u' = 1 \quad v = -\frac{e^{-st}}{s} \quad v' = e^{-st} \\ &= -\frac{e^{-st}}{s} t - \int -\frac{e^{-st}}{s} dt \\ &= \left[-\frac{e^{-st}}{s} t - \frac{e^{-st}}{s^2} \right]_1^{\infty} \Rightarrow 0 \\ &= 0 - \left(-\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} \right) \\ &= e^{-s} \left(\frac{1}{s} + \frac{1}{s^2} \right) \end{aligned}$$

$$\text{b)} \quad y'' + y = 2e^t, \quad y(0) = y'(0) = 0$$

$$\mathcal{L}(y'') = s^2 F(s) - s f(0) - f'(0)$$

$$\mathcal{L}(y) = F(s)$$

$$2 \mathcal{L}(e^t) = \frac{2}{s-1}$$

$$\Rightarrow s^2 F(s) + F(s) = \frac{2}{s-1}$$

$$F(s)(s^2 + 1) = \frac{2}{s-1}$$

$$F(s) = \frac{2}{(s^2 + 1)(s-1)}$$

$$\frac{2}{(s^2 + 1)(s-1)} = \frac{As+B}{s^2 + 1} + \frac{C}{s-1} \Rightarrow 2 = (As+B)(s-1) + C(s^2 + 1) = s^2(A+C) + s(B-A) - B + C$$

$\begin{matrix} A+C=0 & (s^2) \\ B-A=0 & (s) \\ C-B=2 & (1) \end{matrix} \Rightarrow A=B=-1, C=1$

$$F(s) = \frac{-s}{s^2 + 1} - \frac{1}{s^2 + 1} + \frac{1}{s-1}$$

$$= \underline{-\cos(t) - \sin(t) + e^t}$$

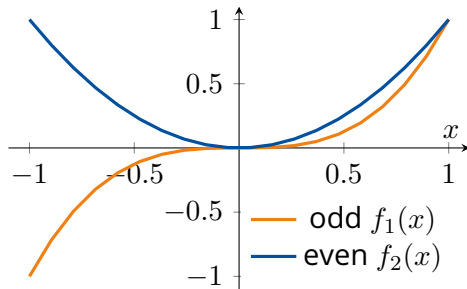
$$\text{c)} \quad F(s) = \frac{1}{s^2 + 2s + 17} = \frac{1}{(s+1)^2 + 4^2} \quad \mathcal{L}(e^{at} f(t)) = F(s-a)$$

$$= \frac{1}{4} \cdot e^{-t} \sin(4t)$$

Reminder. Even and odd functions

A function is called **even** if $f(x) = f(-x)$ holds for all x .

A function is called **odd** if $f(x) = -f(-x)$ holds for all x .



we have the following

f	g	$f \cdot g$
even	even	even
odd	even	odd
even	odd	odd
odd	odd	even

Examples. For $n \in \mathbb{N}$: $\cos(nx)$, is even, $\sin(nx)$ is odd.

Even better.

$\sin^2(nx) \Rightarrow \text{even}$

▶ if f is odd then $\int_{-L}^L f(x) dx = 0$

▶ if f is even then $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$

a) Let

$$f(x) = 1 + x, \quad -\pi < x < \pi.$$

Verify the following complex Fourier series expansion for f

$$1 + \sum_{n \neq 0} \frac{i(-1)^n}{n} e^{inx}.$$

b) Why is

$$f(x) = 1 + \sum_{n \neq 0} \frac{i(-1)^n}{n} e^{inx}$$

for $-\pi < x < \pi$?

c) Compute the Fourier transform of

$$f(x) = \begin{cases} \sin(x) & |x| < 1, \\ 0 & |x| \geq 1. \end{cases}$$

a) $f(x) = 1 + x$

COMPLEX WAY

$$c_0 = a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1+x dx = \frac{1}{2\pi} \left[x + \frac{x^2}{2} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left(\pi + \frac{\pi^2}{2} - (-\pi + \frac{(-\pi)^2}{2}) \right) = 1$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1+x) e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} dx + x e^{-inx} dx$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} dx = \frac{1}{2\pi} \left[\frac{e^{-inx}}{-in} \right]_{-\pi}^{\pi} = \frac{1}{2\pi in} (e^{-in\pi} - e^{in\pi}) = -\frac{1}{\pi n} \left(\frac{e^{in\pi} - e^{-in\pi}}{2i} \right) = -\frac{\sin(n\pi)}{\pi n} = 0, n \in \mathbb{Z}$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} dx = x \left(-\frac{e^{-inx}}{in} \right) + \frac{1}{in} \int_{-\pi}^{\pi} e^{-inx} dx = \frac{1}{2\pi} \left[-\frac{x e^{-inx}}{in} + \frac{e^{-inx}}{n^2} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left(-\frac{\pi e^{-in\pi}}{in} + \frac{e^{-in\pi}}{n^2} - \frac{\pi e^{in\pi}}{in} - \frac{e^{in\pi}}{n^2} \right)$$

$$\frac{1}{2\pi} \left(-\frac{\pi e^{-in\pi}}{in} + \frac{e^{-in\pi}}{n^2} - \frac{\pi e^{in\pi}}{in} - \frac{e^{in\pi}}{n^2} \right) = \frac{1}{2\pi} \left(-\frac{\pi e^{-in\pi}}{in} - \frac{\pi e^{in\pi}}{in} + \frac{(-1)^{n+1}}{n^2} - \frac{(-1)^{n+1}}{n^2} \right) = -\frac{1}{in} \left(\frac{e^{in\pi} + e^{-in\pi}}{2} \right) = -\frac{\cos(n\pi)}{in} = \frac{(-1)^{n+1}}{in}$$

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{\frac{inx}{2}} = \sum_{n=-\infty}^{\infty} \frac{(-1)^{n+1}}{in} e^{\frac{inx}{2}} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{in} e^{inx} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{in} e^{-inx}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{in} (e^{inx} - e^{-inx}) = 1 +$$

$$f(x) = 1 + x$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1+x dx = \frac{1}{2\pi} \left[x + \frac{x^2}{2} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left(\pi + \frac{\pi^2}{2} - (-\pi + \frac{(-\pi)^2}{2}) \right) = 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (1+x) \cos(nx) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(nx) dx + \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(nx) dx = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (1+x) \sin(nx) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) dx + x \sin(nx) dx = \frac{1}{\pi} \left[-\frac{\cos(nx)}{n} - \frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left(-\frac{\cos(n\pi)}{n} - \frac{\pi \cos(n\pi)}{n} - \left(-\frac{\cos(-n\pi)}{n} + \frac{\pi \cos(-n\pi)}{n} \right) \right) = -\frac{2}{\pi} \cdot \frac{\pi \cos(n\pi)}{n} = -\frac{2 \cos(n\pi)}{n}$$

$$c_0 = a_0$$

$$c_n = \frac{1}{2} (a_n - i b_n) \quad n > 0 \Rightarrow c_n = \frac{i \cos(n\pi)}{n}, \quad n > 0$$

$$= \frac{1}{2} (a_n + i b_n) \quad n < 0 \Rightarrow c_n = -\frac{i \cos(n\pi)}{n}, \quad n < 0 = \frac{i \cos(n\pi)}{n} = \frac{i(-1)^n}{n}$$

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{\frac{inx}{2}} \quad \text{eller} \quad f(x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$= 1 + \sum_{n \neq 0} \frac{i(-1)^n}{n} e^{inx}, \quad n \in \mathbb{Z}$$

b) Fordi f er kontinuertlig deriverbar på intervallet

c)

$$f(x) = \sin(x), |x| < 1 \quad f(x) = \sin(x) \quad x \in [-1, 1]$$

$$f(x) = 0, |x| \geq 1 \Rightarrow f(x) = 0 \quad x \in (-\infty, -1) \text{ og } x \in (1, \infty) \Rightarrow \text{Fourier } f(x) = 0$$

$$f(x) = 0, \quad x \geq 1 \quad \text{og} \quad x \leq -1$$

$$f(x) = \sin(x), \quad -1 < x < 1$$

$$\hat{f}(\omega) = F(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \sin(x) e^{-i\omega x} dx, \quad e^{-i\omega x} = \cos(x\omega) - i \sin(x\omega)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \sin(x) (\cos(x\omega) - i \sin(x\omega)) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \sin(x) \cos(x\omega) dx - \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \sin(x) i \sin(x\omega) dx$$

$$= -\frac{i}{\sqrt{2\pi}} \int_{-1}^1 \sin(x) \sin(\omega x) dx, \quad 2 \sin(a) \sin(b) = \cos(a-b) - \cos(a+b)$$

$$= -\frac{i}{\sqrt{2\pi}} \int_0^1 \cos(x-\omega x) - \cos(x+\omega x) dx$$

$$= -\frac{i}{\sqrt{2\pi}} \left[\frac{\sin(x(1-\omega))}{(1-\omega)} - \frac{\sin(x(1+\omega))}{(1+\omega)} \right]_0^1 \Rightarrow 0$$

$$\hat{f}(\omega) = -\frac{i}{\sqrt{2\pi}} \left(\frac{\sin(1-\omega)}{(1-\omega)} - \frac{\sin(1+\omega)}{(1+\omega)} \right)$$

Problem 3 TMA4130 Mathematics 4N: [6 points]

Solve the initial value problem for the wave equation ($u_{tt} = \partial^2 u / \partial t^2$, $u_{xx} = \partial^2 u / \partial x^2$, $u_t = \partial u / \partial t$ mean partial derivatives)

$$u_{tt} = u_{xx}, \quad u(x, 0) = \sin(x), \quad u_t(x, 0) = e^x,$$

using d'Alembert's solution.

$$\text{Given: } \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$\text{Solution by d'Alembert's is: } u(x, t) = \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x}$$

$$u(x, t) = \frac{1}{2} (\sin(x+t) + \sin(x-t)) + \int_{x-t}^{x+t} e^{\tilde{x}} d\tilde{x}$$

$$\sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

$$= \sin(x) \cos(t) + \frac{1}{2} [e^{\tilde{x}}]_{x-t}^{x+t}$$

$$= \sin(x) \cos(t) + \frac{1}{2} (e^{x+t} - e^{x-t}) \quad \text{OK!}$$

Problem 4 [20 points]

Consider the following heat equation

$$u_t(x, t) = \frac{1}{2} u_{xx}(x, t), \quad t \geq 0, \quad 0 \leq x \leq \pi,$$

with boundary conditions

$$\Rightarrow c = \frac{1}{2}$$

$$u(0, t) = u(\pi, t) = 0, \quad t \geq 0;$$

and the initial condition

$$u(x, 0) = x(\pi - x), \quad 0 \leq x \leq \pi.$$

a) Find the Fourier sine series solution of the above heat equation by using the separation of variables method.

b) Let M, N be two natural numbers, and define $h = \pi/M$ and $k = 1/N$. Introduce $x_i = ih$ for $i = 0, \dots, M$ and $t_n = nk$ for $n = 0, 1, 2, \dots$. Write down an explicit difference scheme (based on finite differences and (forward) Euler's method) for $U_i^n \approx u(x_i, t_n)$.

c) Let $M = 4$ and $N = 20$, and compute the approximate solution for $u(\pi/4, 0.1)$.

Lösningssformel för gittarsträng

Varmelikningen

$$u_t(x, t) = c^2 u_{xx}(x, t),$$

med randkrav

$$u(0, t) = u(L, t) = 0,$$

og initialkrav

$$u(x, 0) = f(x)$$

löses av

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{c n \pi}{L}\right)^2 t} \sin \frac{n \pi}{L} x,$$

der

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n \pi}{L} x dx.$$

$$a) \quad u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{1}{2} \frac{n\pi}{a}\right)^2 t} \sin\left(\frac{n\pi}{a} x\right) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2 \pi^2}{2} t} \sin(n\pi x)$$

$$A_n = \frac{2}{a} \int_0^a (\pi x - x^2) \sin(n\pi x) dx$$

$$= \frac{2}{a} \int_0^a x \sin(n\pi x) dx - \frac{2}{a} \int_0^a x^2 \sin(n\pi x) dx$$

$$\frac{2}{a} \int_0^a x \sin(n\pi x) dx = \frac{2}{a} \left[-\frac{x \cos(n\pi x)}{n} + \frac{1}{n} \int \cos(n\pi x) dx \right]_0^a = \frac{2}{a} \left[-\frac{x \cos(n\pi x)}{n} + \frac{\sin(n\pi x)}{n^2} \right]_0^a \Rightarrow 0, n \in \mathbb{Z}$$

$$= \frac{2}{a} \left(-\frac{\pi \cos(\pi n)}{n} + 0 - (0 - 0) \right)$$

$$= -\frac{2\pi}{a} \frac{(-1)^n}{n}$$

$$\frac{2}{a} \int_0^a x^2 \sin(n\pi x) dx = \frac{2}{a} \left[-\frac{x^2 \cos(n\pi x)}{n} + \frac{1}{n} \int x \cos(n\pi x) dx \right]_0^a$$

$$= \frac{2}{a} \left[-\frac{x^2 \cos(n\pi x)}{n} + \frac{1}{n} \left(\frac{x \sin(n\pi x)}{n} - \frac{1}{n} \int \sin(n\pi x) dx \right) \right]_0^a$$

$$= \frac{2}{a} \left[-\frac{x^2 \cos(n\pi x)}{n} + \frac{x \sin(n\pi x)}{n^2} + \frac{\cos(n\pi x)}{n^3} \right]_0^a$$

$$= \frac{2}{a} \left(-\frac{\pi^2 \cos(\pi n)}{n} + 0 + \frac{\cos(\pi n)}{n^3} - (0 + 0 + \frac{1}{n^3}) \right)$$

$$= \frac{2}{a} \left(-\frac{\pi^2 \cos(\pi n)}{n} + \frac{\cos(\pi n)}{n^3} - \frac{1}{n^3} \right)$$

$$= -\frac{2\pi}{a} \frac{(-1)^n}{n} + \frac{2(-1)^n}{\pi n^3} - \frac{2}{\pi n^3}$$

$$\frac{2}{a} \int_0^a x \sin(n\pi x) dx - \frac{2}{a} \int_0^a x^2 \sin(n\pi x) dx = \frac{2}{a} \left(-\frac{2\pi}{a} \frac{(-1)^n}{n} - \left(-\frac{2\pi}{a} \frac{(-1)^n}{n} + \frac{2(-1)^n}{\pi n^3} - \frac{2}{\pi n^3} \right) \right) = \frac{4}{\pi n^3} (1 - (-1)^n)$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{4}{\pi n^3} (1 - (-1)^n) e^{-\frac{n^2 \pi^2}{2} t} \sin(n\pi x) \quad \text{OK!}$$

b & c) Paper OK!

Problem 5 [10 points]

Find a, b, c, d such that the polynomial

$$p(x) = ax^3 + bx^2 + cx + d$$

interpolates the points

	x_0	x_1	x_2	x_3
x_i	0	2	3	4
y_i	1	5	10	17
	y_0	y_1	y_2	y_3

$$L_j(x) = \prod_{\substack{n=0 \\ n \neq j}}^3 \frac{x - x_n}{x_j - x_n}$$

$$L(x) = \sum_{j=0}^3 L_j y_j$$

$$L_0 = \left(\frac{x - x_1}{x_0 - x_1} \right) \left(\frac{x - x_2}{x_0 - x_2} \right) \left(\frac{x - x_3}{x_0 - x_3} \right) = \left(\frac{x - 2}{0 - 2} \right) \left(\frac{x - 3}{0 - 3} \right) \left(\frac{x - 4}{0 - 4} \right) = -\frac{1}{24} (x^3 - 9x^2 + 26x - 24)$$

$$L_1 = \left(\frac{x - x_0}{x_1 - x_0} \right) \left(\frac{x - x_2}{x_1 - x_2} \right) \left(\frac{x - x_3}{x_1 - x_3} \right) = \left(\frac{x - 0}{2 - 0} \right) \left(\frac{x - 3}{2 - 3} \right) \left(\frac{x - 4}{2 - 4} \right) = -\frac{1}{4} (x^3 - 7x^2 + 12x)$$

$$L_2 = \left(\frac{x - x_0}{x_2 - x_0} \right) \left(\frac{x - x_1}{x_2 - x_1} \right) \left(\frac{x - x_3}{x_2 - x_3} \right) = \left(\frac{x - 0}{3 - 0} \right) \left(\frac{x - 2}{3 - 2} \right) \left(\frac{x - 4}{3 - 4} \right) = -\frac{1}{3} (x^3 - 6x^2 + 8x)$$

$$L_3 = \left(\frac{x - x_0}{x_3 - x_0} \right) \left(\frac{x - x_1}{x_3 - x_1} \right) \left(\frac{x - x_2}{x_3 - x_2} \right) = \left(\frac{x - 0}{4 - 0} \right) \left(\frac{x - 2}{4 - 2} \right) \left(\frac{x - 3}{4 - 3} \right) = \frac{1}{8} (x^3 - 5x^2 + 6x)$$

$$L(x) = \sum L_j y_j = -\frac{1}{24} (x^3 - 9x^2 + 26x - 24) - \frac{5}{4} (x^3 - 7x^2 + 12x) - \frac{10}{3} (x^3 - 6x^2 + 8x) + \frac{17}{8} (x^3 - 5x^2 + 6x)$$

$$= -x^3 \left(\frac{1}{24} + \frac{5}{4} + \frac{10}{3} - \frac{17}{8} \right) + x^2 \left(\frac{9}{24} + \frac{35}{4} + 20 - \frac{85}{8} \right) - x \left(\frac{26}{24} + 15 + \frac{80}{3} - \frac{102}{8} \right) + 1$$

$$L(x) = -\frac{5}{2}x^3 + \frac{37}{2}x^2 - 30x + 1 \quad \text{svaret skulle bli } p(x) = x^2 + 1 \quad \text{☹}$$

$$\Rightarrow a = -\frac{5}{2}, \quad b = \frac{37}{2}, \quad c = -30, \quad d = 1$$

Newton's

$$[y_0] = y_0 \quad [y_1] = y_1 \quad [y_2] = y_2 \quad [y_3] = y_3$$

$$[y_0, y_1] = \frac{y_1 - y_0}{x_1 - x_0} \quad [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1} \quad [y_2, y_3] = \frac{y_3 - y_2}{x_3 - x_2}$$

$$[y_0, y_1, y_2] = \frac{[y_1, y_2] - [y_0, y_1]}{x_2 - x_0} \quad [y_1, y_2, y_3] = \frac{[y_2, y_3] - [y_1, y_2]}{x_3 - x_1}$$

$$[y_0, y_1, y_2, y_3] = \frac{[y_1, y_2, y_3] - [y_0, y_1, y_2]}{x_3 - x_0}$$

$$[y_0] = 1 \quad [y_1] = 5 \quad [y_2] = 10 \quad [y_3] = 17$$

$$[y_0, y_1] = \frac{5-1}{2-0} = 2 \quad [y_1, y_2] = \frac{10-5}{3-2} = 5 \quad [y_2, y_3] = \frac{17-10}{4-3} = 7$$

$$[y_0, y_1, y_2] = \frac{5-2}{3-0} = 1 \quad [y_1, y_2, y_3] = \frac{7-5}{4-2} = 1$$

$$[y_0, y_1, y_2, y_3] = \frac{[y_1, y_2, y_3] - [y_0, y_1, y_2]}{x_3 - x_0} = \frac{1-1}{17-1} = 0$$

$$N(x) = 1 + 2(x-0) + 1(x-0)(x-2) + 0$$

$$= \underline{x^2 + 1} \quad \Rightarrow a = c = 0, \quad b = d = 1 \quad \text{OK!}$$

Problem 6 [10 points]

The integral

$$\int_0^1 f(x) dx,$$

can be approximated by the Simpson formula

$$S = \frac{1}{6} \left(f(0) + 4f(0.5) + f(1) \right).$$

a) Apply the Simpson formula to the integral

$$\int_0^1 x^3 dx.$$

b) Determine the degree of precision for the Simpson formula.

$$a) \quad S = \frac{1}{6} (f(0) + 4f(0.5) + f(1)) \quad , \quad f(x) = x^3$$

$$S = \frac{1}{6} (0 + 4(0.5)^3 + 1^3) = \frac{4}{48} + \frac{1}{6} = \frac{12}{48} = \underline{0.25} = \int_0^1 x^3 dx$$

$$b) \quad \overset{\text{Integralen}}{I[x^j]}(a, b) = \overset{\text{kvadraturen}}{Q[x^j]}(a, b) \quad \text{Præsjønsgraden er høyeste j der dette stemmer}$$

$$\int_0^1 x^4 dx = \left[\frac{x^5}{5} \right]_0^1 = 0.2 \neq \frac{1}{6} (0 + 4(0.5)^4 + 1^4) = \frac{1}{24} + \frac{1}{6} = \frac{5}{24} = 0.21 \Rightarrow \text{Præsjønsgr. 3}$$

Problem 7 [10 points]

Let r be the solution of the following equation

$$x + \ln(x - 1) = 0, \quad 1 < x < 2.$$

Show that the solution is unique. Starting from

$$x_0 = 1.25,$$

apply one step of Newton's iteration, and compute x_1 .

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$f(x) = x - \ln(x-1) \Rightarrow f'(x) = 1 - \frac{1}{x-1}$$

$$f(1.25) = 1.25 - \ln(1.25-1) = \frac{5}{4} + 2\ln(2) = 2.636 \quad \& \quad f'(1.25) = 1 - \frac{1}{1.25-1} = -3$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.25 - \frac{2.636}{-3} = 0.37$$

Problem 8 [10 points]

Heun's method is given by:

$$k_1 = f(x_n, y_n), \quad = y'(x_n)$$

$$k_2 = f(x_n + h, y_n + hk_1),$$

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2).$$

- a) Apply one step with step size $h = 0.1$ using the above method on the problem:

$$y' = -2xy, \quad y(0) = 1.$$

Find the exact solution of the above equation and compute the error.

- b) Find the stability function $R(z)$ for Heun's method. Find also the corresponding stability interval.

$$a) \quad k_1 = f(x_n, y(x_n)) = y'(0, 1) = -2 \cdot 0 \cdot 1 = 0$$

$$k_2 = f(0 + 0.1, 1 + 0.1 \cdot 0) = y'(0.1, 1) = -2 \cdot 0.1 \cdot 1 = -0.2$$

$$y_1 = 1 + \frac{0.1}{2}(0 + (-0.2)) = 1 - \frac{0.04}{2} = 0.99$$

$$y' = -2xy$$

$$\int \frac{1}{y} dy = -\int 2x dx$$

$$\ln(y) = -x^2$$

$$y = e^{-x^2} \quad \text{Es sagt Lösung}$$

$$\text{Feil:} \quad e = y(x_{n+1}) - y_{n+1} = e^{-0.1^2} - 0.99 \quad \text{OK!}$$

$$b) \quad \text{Ingen peiling}$$